Lecture 3

Sequence limits and their algebra

Uses.

Summary 1: Negating a quantified statement reverses each quantifier. The negation of $P \Rightarrow Q$ is P and not Q.
Summary 2: A function assigns one value in its codomain to each input in its domain. For $h \geq -1$ and $n \in \mathbb{N}$, Bernoulli's inequality gives $(1+h)^n \geq 1+nh$. The triangle estimates give $|x+y| \leq |x|+|y|$ and $\big||x|-|y|\big| \leq |x-y|$. For each $x \in \mathbb{R}$, the Archimedean property supplies $N \in \mathbb{N}$ with $N > x$.

Objectives. *We give a precise meaning to convergence of a sequence. An error estimate will turn a requested accuracy into an index after which every term has that accuracy. We use these estimates to prove the laws for sums, products and quotients of convergent sequences.*

By the end of this lecture you will be able to:

- State the definition of convergence and its negation.
- Choose an index and verify an explicit sequence limit for every later term.
- Prove that a convergent real sequence has a unique limit.
- Apply the sum, product and quotient laws with their hypotheses.

3.1 The ε–N definition

A *real sequence* is a function from $\mathbb{N}$ to $\mathbb{R}$. We write its value at $n \in \mathbb{N}$ as a_n and call it the *nth term*. The sequence (a_n) records the terms in their order. For example, the sequence given by $a_n = 1/n$ begins with 1, $1/2$, $1/3$ and $1/4$.

The terms of this sequence become close to zero. A convergence statement must say how close they become and whether they stay close. We measure the error relative to a proposed limit $a \in \mathbb{R}$ by $|a_n - a|$. An accuracy $\varepsilon > 0$ specifies the open interval

$$(a-\varepsilon, a+\varepsilon).$$

A term lies in this interval exactly when its error is less than ε.

Definition 3.1 (Convergence). A real sequence (a_n) *converges* to $a \in \mathbb{R}$ if

$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N} \quad \forall n \in \mathbb{N} : \quad n \geq N \implies |a_n - a| < \varepsilon. \tag{3.1}$$

We write $a_n \to a$. A sequence is *convergent* if it converges to some $a \in \mathbb{R}$. A sequence that is not convergent is *divergent*.

For $a_n = 1/n$, an accuracy of $\varepsilon = 1/4$ is met by every term with $n \geq 5$. Figure 3.1 shows this choice of N. The term at $n = 4$ lies on the boundary. It fails the strict error condition.

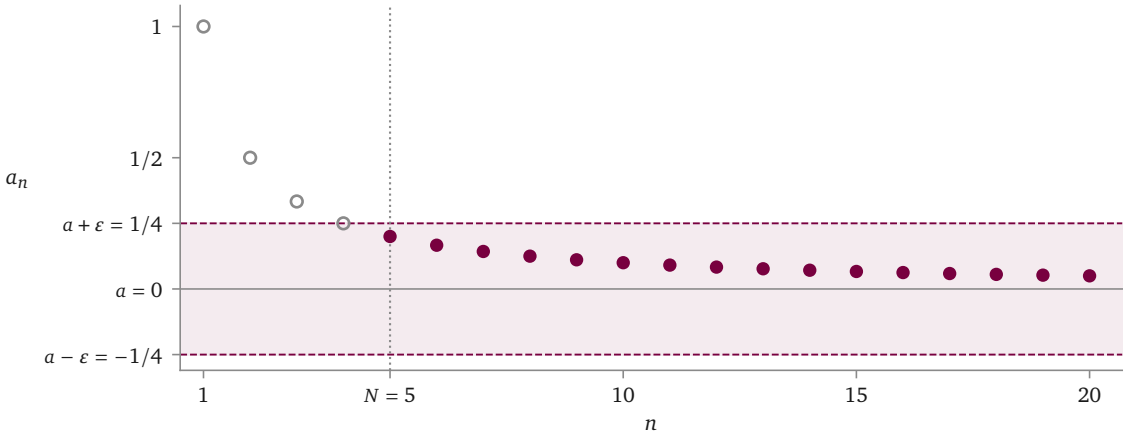


Figure 3.1: The sequence $a_n = 1/n$ and the band $|a_n| < 1/4$. Every filled point has $n \geq N = 5$ and lies strictly inside the band.

A fixed band tests only one accuracy. Convergence requires the same kind of statement for every $\varepsilon > 0$. A smaller ε may require a larger N. The index N may depend on ε and on the sequence. It must be fixed before the index $n \geq N$ is chosen.

We say that a property holds *eventually* if it holds for every $n \geq N$ for some $N \in \mathbb{N}$. Thus convergence to a means that each positive error tolerance is met eventually. The earlier terms need not meet it.

Negating the definition

The negation rules of Summary 1 give an exact test for failure of convergence to a *specified* $a \in \mathbb{R}$:

$$\exists \varepsilon_0 > 0 \quad \forall N \in \mathbb{N} \quad \exists n \in \mathbb{N} : \quad n \geq N \text{ and } |a_n - a| \geq \varepsilon_0. \tag{3.2}$$

Negating the implication gives $n \geq N$ together with failure of the error condition. Negating the strict error bound gives $|a_n - a| \geq \varepsilon_0$.

The tolerance ε_0 is fixed. Every proposed cutoff has at least one term with index $n \geq N$ that fails this tolerance. Those failing terms may depend on N. They need not be all the later terms.

For example, the sequence given by $a_n = (-1)^n$ does not converge to zero. Take $\varepsilon_0 = 1$. For any $N \in \mathbb{N}$, the choice $n = N$ gives $|a_n - 0| = 1$. This disproves convergence to zero. To prove that the sequence is divergent, we must rule out every proposed $a \in \mathbb{R}$. We will do this in § 3.3.

3.2 Three ε–N proofs

An explicit proof starts with a given $\varepsilon > 0$. We estimate the error in terms of n, choose N, and then check every $n \geq N$. The Archimedean property in Summary 2 lets us choose an integer strictly above any real bound. A convenient cutoff is enough. The smallest possible cutoff is not required.

Example 3.2 (Reciprocals). The sequence given by $a_n = 1/n$ converges to zero. Its error is

$$|a_n - 0| = \frac{1}{n}.$$

For a given $\varepsilon > 0$, choose $N \in \mathbb{N}$ with $N > 1/\varepsilon$. Then check every $n \geq N$:

$$|a_n - 0| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

This proves $1/n \to 0$ by Definition 3.1.

The last line checks an entire infinite sequence of inequalities at once. It uses $n \geq N$ to replace the variable error by the fixed bound $1/N$. For instance, if $\varepsilon = 1/10$, we may take $N = 11$. Taking $N = 10$ would fail at $n = 10$, where the error equals the tolerance.

Example 3.3 (A rational expression). We prove the limit

$$a_n = \frac{2n+1}{n+3} \longrightarrow 2.$$

Subtracting the proposed limit gives the error estimate

$$|a_n - 2| = \left|\frac{2n+1-2(n+3)}{n+3}\right| = \frac{5}{n+3} \leq \frac{5}{n}.$$

For a given $\varepsilon > 0$, choose $N \in \mathbb{N}$ with $N > 5/\varepsilon$. Then check every $n \geq N$:

$$|a_n - 2| = \frac{5}{n+3} \leq \frac{5}{n} \leq \frac{5}{N} < \varepsilon.$$

This proves the stated limit by Definition 3.1.

For $\varepsilon = 1/100$, the exact error condition is $n + 3 > 500$. The smallest cutoff is $N = 498$. The proof's simpler choice $N = 501$ also works. Both choices control every later term.

Example 3.4 (Geometric decay). Fix $q \in \mathbb{R}$ with $|q| < 1$. We prove

$$q^n \longrightarrow 0.$$

If $q = 0$, every term is zero and $N = 1$ works for every $\varepsilon > 0$. Suppose now that $0 < |q| < 1$. Put $r = |q|$ and define the positive number

$$h = \frac{1}{r} - 1.$$

Then $r = 1/(1+h)$. Bernoulli's inequality from Summary 2 gives $(1+h)^n \geq 1 + nh$ for every $n \in \mathbb{N}$. Taking reciprocals of these positive quantities gives

$$|q^n| = r^n = \frac{1}{(1+h)^n} \leq \frac{1}{1+nh} < \frac{1}{nh}.$$

For a given $\varepsilon > 0$, choose $N \in \mathbb{N}$ with $N > 1/(h\varepsilon)$. Then check every $n \geq N$:

$$|q^n - 0| \leq \frac{1}{1+nh} < \frac{1}{nh} \leq \frac{1}{Nh} < \varepsilon.$$

This proves the stated limit by Definition 3.1, including negative values of q with $|q| < 1$.

3.3 Uniqueness and boundedness

A sequence (a_n) is *bounded* if there is $M > 0$ such that

$$|a_n| \le M \qquad \text{for every } n \in \mathbb{N}.$$

The bound controls all terms, including any early exceptions to an error tolerance. Boundedness does not specify a limit.

Theorem 3.5 (Uniqueness and boundedness)**.** *Every convergent real sequence has a unique limit and is bounded.*

Proof of uniqueness. We argue by contradiction. Suppose $a_n \to a$ and $a_n \to b$, where $a \neq b$. Define $d = |a - b| > 0$. Convergence to a supplies $N_a \in \mathbb{N}$ such that $|a_n - a| < d/3$ for all $n \ge N_a$. Convergence to b supplies $N_b \in \mathbb{N}$ such that $|a_n - b| < d/3$ for all $n \ge N_b$.

Choose $n \ge \max\{N_a, N_b\}$. Both estimates hold at this same index. By the triangle inequality from Summary 2,

$$d = |a - b| \le |a - a_n| + |a_n - b| < \frac{d}{3} + \frac{d}{3} = \frac{2d}{3}.$$

This contradicts $d > 0$. Hence $a = b$. ∎

We may therefore write the limit as $\lim_{n\to\infty} a_n = a$.

Proof of boundedness. Suppose $a_n \to a$. With $\varepsilon = 1$, the definition supplies $N \in \mathbb{N}$ such that $|a_n - a| < 1$ for every $n \ge N$. The triangle inequality gives

$$|a_n| \le |a_n - a| + |a| < |a| + 1 \qquad (n \ge N).$$

Define the positive number

$$M = \max\{|a| + 1, |a_1|, \ldots, |a_N|\}.$$

This finite maximum exists and includes every term with index at most N. For $n > N$, the preceding estimate gives $|a_n| < |a| + 1 \le M$. Thus $|a_n| \le M$ for every $n \in \mathbb{N}$. ∎

The tolerance 1 supplies a bound for the entire later part of the sequence. It does not need to be arbitrarily small for this purpose. The finite maximum includes the earlier terms, which convergence alone does not control. A bound on only the later terms would not yet be the bound asserted in the theorem.

Boundedness does not imply convergence

The sequence given by $a_n = (-1)^n$ is bounded, since $|a_n| = 1$. We show that it has no limit. Suppose it converged to some $a \in \mathbb{R}$. With $\varepsilon = 1$, convergence would give $N \in \mathbb{N}$ such that $|a_n - a| < 1$ for all $n \ge N$.

Choose an even index $m \ge N$ and an odd index $k \ge N$. These choices are possible after every cutoff. Then $a_m = 1$ and $a_k = -1$, so both $|1 - a| < 1$ and $|a - (-1)| < 1$. The triangle inequality would give

$$2 = |1 - (-1)| \le |1 - a| + |a - (-1)| < 1 + 1 = 2.$$

This contradiction rules out every $a \in \mathbb{R}$. Thus a bounded sequence may be divergent.

Small successive differences do not imply convergence

Consider the sequence given by $b_n = \sqrt{n}$. Its successive difference is

$$c_n = b_{n+1} - b_n = \sqrt{n+1} - \sqrt{n}.$$

Rationalising gives the estimate

$$0 < c_n = \frac{1}{\sqrt{n+1} + \sqrt{n}} < \frac{1}{\sqrt{n}}.$$

For a given $\varepsilon > 0$, choose $N \in \mathbb{N}$ with $N > 1/\varepsilon^2$. Then check every $n \geq N$. Since $n \geq N > 1/\varepsilon^2$, taking positive square roots gives $\sqrt{n} > 1/\varepsilon$. Hence

$$|c_n - 0| < \frac{1}{\sqrt{n}} < \varepsilon.$$

Thus the successive differences tend to zero.

The sequence (b_n) is nevertheless unbounded. Given $M > 0$, the Archimedean property supplies $n \in \mathbb{N}$ with $n > M^2$. Then $b_n = \sqrt{n} > M$. By Theorem 3.5, the sequence (b_n) cannot converge to a real limit. Small changes from one term to the next therefore do not establish convergence of the terms themselves.

3.4 The algebra of limits

A direct error proof establishes a limit for one sequence. To combine sequences, we need to control several errors at the same indices. If one estimate holds for $n \geq N_1$ and another for $n \geq N_2$, both hold for every $n \geq \max\{N_1, N_2\}$. Their cutoffs may differ. The maximum gives a single cutoff for the combined argument.

Changing finitely many terms does not change a limit. To check this, fix $\varepsilon > 0$ and take an original error cutoff. Increase it past all changed indices. Every remaining error is unchanged and is less than ε. This observation also lets us fill any finitely many undefined initial positions of a formula with arbitrary real values. The convergence assertion concerns the resulting sequence.

Theorem 3.6 (Limit laws). *Let $a_n \to a$ and $b_n \to b$ be real sequences, and let $c \in \mathbb{R}$. Then*

$$a_n + b_n \to a + b, \qquad ca_n \to ca, \qquad a_n b_n \to ab.$$

If $b \neq 0$, then $b_n \neq 0$ eventually and

$$\frac{a_n}{b_n} \longrightarrow \frac{a}{b}.$$

Undefined initial quotients may be assigned arbitrary real values.

Proof of the sum law. Let $\varepsilon > 0$ be given. Choose $N_a \in \mathbb{N}$ such that $|a_n - a| < \varepsilon/2$ for every $n \geq N_a$. Choose $N_b \in \mathbb{N}$ such that $|b_n - b| < \varepsilon/2$ for every $n \geq N_b$. Put $N = \max\{N_a, N_b\}$. Then check every $n \geq N$:

$$|(a_n + b_n) - (a + b)| \leq |a_n - a| + |b_n - b| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

This is convergence of the sum to $a + b$. ∎

Each summand is controlled at the smaller tolerance $\varepsilon/2$. Using tolerance ε for each would give only a total error below 2ε. The definition allows a choice at any positive tolerance, so it allows the two choices used in the proof.

Proof of the product law. By Theorem 3.5, choose $M > 0$ with $|a_n| \le M$ for every $n \in \mathbb{N}$. Subtracting the proposed product limit gives the identity

$$a_n b_n - ab = a_n(b_n - b) + b(a_n - a).$$

This writes the error as a sum involving the two known errors. Let $\varepsilon > 0$ be given and put $\delta = \varepsilon/(M + |b| + 1) > 0$. Choose $N \in \mathbb{N}$ such that both $|a_n - a| < \delta$ and $|b_n - b| < \delta$ hold for every $n \ge N$. Such an N is the maximum of the two convergence cutoffs. Then check every $n \ge N$:

$$|a_n b_n - ab| \le M|b_n - b| + |b||a_n - a| \le (M + |b|)\delta < \varepsilon.$$

The last inequality uses $M + |b| < M + |b| + 1$. This proves the product law. ∎

The bound M controls the coefficient a_n in the error identity. Without that control, making $|b_n - b|$ small would not yet control $a_n(b_n - b)$. The boundedness theorem supplies this missing step. The extra 1 in the denominator of δ makes the final comparison strict.

Every constant sequence with value c converges to c, since its error is zero. Applying the product law to a_n and that constant sequence proves the scalar-multiple law. It also covers $c = 0$. Replacing c by -1 and using the sum law gives the corresponding law for differences.

Proof of the quotient law. Suppose $b \ne 0$. Convergence with tolerance $|b|/2$ supplies $N_0 \in \mathbb{N}$ such that $|b_n - b| < |b|/2$ for every $n \ge N_0$. The reverse triangle inequality from Summary 2 gives

$$|b_n| \ge |b| - |b_n - b| > \frac{|b|}{2} > 0 \qquad (n \ge N_0).$$

Thus the later denominators are nonzero.

Let $\varepsilon > 0$ be given. In addition, choose a convergence cutoff for $|b_n - b| < \varepsilon|b|^2/2$. Take N to be the maximum of that cutoff and N_0. Then check every $n \ge N$:

$$\left|\frac{1}{b_n} - \frac{1}{b}\right| = \frac{|b_n - b|}{|b_n||b|} \le \frac{2}{|b|^2}|b_n - b| < \varepsilon.$$

This proves convergence of the reciprocal to $1/b$, after filling any undefined initial terms. The product law applied to a_n and $1/b_n$ now gives $a_n/b_n \to a/b$. ∎

The first cutoff keeps the denominator away from zero. The second controls the requested reciprocal error. A denominator can have a nonzero limit without being nonzero at every early index. Its finitely many early zeros do not enter the proof after the two cutoffs have been combined.

Example 3.7 (A limit computed by the laws)**.** We compute the limit of the sequence

$$u_n = \frac{3n^2 - 2n + 1}{n^2 + 4}.$$

Divide numerator and denominator by n^2, which is positive:

$$u_n = \frac{3 - 2/n + 1/n^2}{1 + 4/n^2}.$$

The limit $1/n \to 0$ was proved directly. Its product with itself gives $1/n^2 \to 0$. The scalar and sum laws therefore give limits 3 and 1 for the normalized numerator and denominator. Since the denominator limit is nonzero, the quotient law applies and gives $u_n \to 3$.

The condition $b \neq 0$ cannot be omitted from the quotient law. For example, both $1/n$ and $1/n^2$ tend to zero. Nevertheless,

$$\frac{1/n}{1/n} = 1, \qquad \frac{1/n^2}{1/n} = \frac{1}{n} \longrightarrow 0.$$

Vanishing numerator and denominator limits do not determine a quotient limit. They also do not prove divergence. The actual ratio still needs an argument.

Summary 3.

- Definition 3.1: for every $\varepsilon > 0$, choose $N \in \mathbb{N}$ so that $|a_n - a| < \varepsilon$ for every $n \geq N$.
- Equation (3.2): failure to converge to a specified $a \in \mathbb{R}$ means that some fixed $\varepsilon_0 > 0$ fails at a term with index $n \geq N$ for every $N \in \mathbb{N}$.
- The limits $1/n \to 0$, $(2n+1)/(n+3) \to 2$ and $q^n \to 0$ for real $|q| < 1$ follow from explicit error bounds.
- A property holds eventually if it holds for all $n \geq N$ for some $N \in \mathbb{N}$. A real sequence is bounded if $|a_n| \leq M$ for some $M > 0$ and every $n \in \mathbb{N}$.
- Theorem 3.5: a convergent real sequence has a unique limit and is bounded. Both assertions are proved here.
- Theorem 3.6: sums, scalar multiples and products preserve limits. Quotients preserve limits when the denominator limit is nonzero.

Boundedness alone does not imply convergence. Neither do successive differences tending to zero. The limit laws use one cutoff for several errors. A quotient requires a nonzero denominator limit. Lecture 4 will use order and completeness to prove existence of further limits. Sheet 3 practises direct limit proofs, subsequences and geometric sums.

References

- R. G. Bartle and D. R. Sherbert, *Introduction to Real Analysis*, 4th ed., Wiley, 2011, §§3.1–3.2.
- S. R. Ghorpade and B. V. Limaye, *A Course in Calculus and Real Analysis*, 2nd ed., Springer, 2018, §2.1, pp. 41–44.